DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Summer Supplementary Examination – 2022 (Pattern 2019)

Branch : Electronics and Tele Communication Eng.

Course: B. Tech. **Semester : V**

Subject Code & Name: BTEXC505 Microcontroller and its applications

Max Marks: 60 **Date:** **Duration: 3.00 Hr.**

Instructions to the Students:

1. *Attempt any five questions.*
2. *Use of non-programmable scientific calculators is allowed.*
3. *Assume suitable data wherever necessary and mention it clearly.*

			Marks
Q. 1	**Solve the following.**		
A)	**Draw architecture of 8051 microcontroller and explain any three blocks of it.**		**6**
B)	**Draw pin diagram of 8051 microcontroller and explain any four pins of it.**		**6**
Q.2	**Solve any two of the following.**		
A)	**Explain any three addressing modes of 8051 microcontroller with suitable examples of each.**		**6**
B)	**Explain following instructions with suitable examples of each.** ***MOV A, @R1*** ***PUSH 23H*** ***SETB 30H***		**6**
C)	**Explain PSW register of 8051 microcontroller.**		**6**
Q. 3	**Solve any two of the following.**		
A)	**Explain TMOD register of 8051 microcontroller.**		**6**
B)	**Assume that 5 numbers are stored from 1000H to 1004H code memory locations. Write an assembly language program to add these numbers and store result at 20H internal RAM location.**		**6**
C)	**Explain SCON register of 8051 microcontroller.**		**6**
Q.4	**Solve any two of the following.**		
A)	**Write a C program for 8051 to transfer the letter “A” serially at 4800 baud continuously. Use 8-bit data and 1 stop bit.**		**6**
B)	**Assuming that we are programming the timers for mode 2, find the value (in hex) loaded into TH for each of the following cases.** ***MOV TH1,#-200; MOV TH0,#-60; MOV TH1,#-3***		**6**

C)	**Explain TCON register of 8051 microcontroller.**		**6**
Q. 5	**Solve the following.**		
A)	**The data pins of an LCD are connected to P1. The information is latched into the LCD whenever its Enable pin goes from high to low. Write an 8051 C program to send "The Earth is but One Country" to this LCD.**		**6**
B)	**Explain following instructions of PIC microcontroller with suitable examples of each.** ***ADDLW*** ***BSF*** ***CLRW***		**6**
Q. 6	**Solve the following.**		
A)	**Explain in brief about data memory map of PIC18.**		**6**
B)	**Explain all 9 pins of RS232 DB9 connector.**		**6**
	***** End *****		